BPS-17 Lecturer Mathematics
AJKPSC / FPSC / PPSC / KPSC
Comprehensive Study Plan

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Metric Spaces & Topology · Functional Analysis
Linear Algebra · Numerical Analysis

Topics Covered:

- Metric Spaces: Definitions & Examples
- Interior, Exterior & Frontier of a Set
- Topological Spaces
- Gram–Schmidt Orthonormalization
- Matrix Algebra & Echelon Forms
- Gauss–Seidel & Jacobi Methods
- Eigenvalues & Eigenvectors
- Open & Closed Spheres
- Sequences & Convergence in Metric Spaces
- Normed & Inner Product Spaces
- Linear Transformations
- Systems of Linear Equations
- Properties of Determinants
- Diagonalization of Symmetric Matrices

Each section: **Concept Summary → Key Formulas → MCQs with Full Explanations**

Reference Material — PSC 2021–26

Contents

1. Metric Spaces — Definitions, Examples & Open/Closed Spheres

A **metric space** (X, d) is a set X with a function $d : X \times X \to [0, \infty)$ satisfying for all $x, y, z \in X$:

1. **Non-negativity:** $d(x, y) \geq 0$, and $d(x, y) = 0 \Leftrightarrow x = y$
2. **Symmetry:** $d(x, y) = d(y, x)$
3. **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$

Open sphere / ball: $B(x_0, r) = \{x \in X : d(x, x_0) < r\}$
Closed sphere: $\bar{B}(x_0, r) = \{x \in X : d(x, x_0) \leq r\}$

Common metrics on $\mathbb{R}^n$:

- Euclidean: $d_2(x, y) = \sqrt{\sum (x_i - y_i)^2}$
- Taxicab: $d_1(x, y) = \sum |x_i - y_i|$
- Chebyshev: $d_\infty(x, y) = \max |x_i - y_i|$
- Discrete: $d(x, y) = 0$ if $x = y$, $d(x, y) = 1$ if $x \neq y$

Key Formulas & Results

Interior, Exterior and Frontier of a set $A \subseteq X$:

$$\text{Int}(A) = A^\circ = \{x \in A : \exists\, r > 0,\ B(x, r) \subseteq A\}$$

$$\text{Ext}(A) = \text{Int}(A^c), \qquad \partial A = \overline{A} \cap \overline{A^c} \quad \text{(Frontier/Boundary)}$$

$$\text{Closure: } \overline{A} = A^\circ \cup \partial A, \qquad X = \text{Int}(A) \cup \partial A \cup \text{Ext}(A) \text{ (disjoint)}$$

1.1. MCQs — Metric Spaces

Q1. Which of the following is *not* a metric on $\mathbb{R}$?

(A) $d(x, y) = |x - y|$

(B) $d(x, y) = |x - y|^2$

(C) $d(x, y) = \dfrac{|x - y|}{1 + |x - y|}$

(D) $d(x, y) = \sqrt{|x - y|}$

Answer: (B) $d(x, y) = |x - y|^2$ **fails** the triangle inequality. Take $x = 0, y = 2, z = 1$: $d(0, 2) = 4$, but $d(0, 1) + d(1, 2) = 1 + 1 = 2 < 4$. Options (A), (C), (D) are all valid metrics.

Q2. In the discrete metric space (X, d), the open ball $B(x_0, 1)$ equals

(A) X

(B) $\{x_0\}$

(C) $\emptyset$

(D) $X \setminus \{x_0\}$

Answer: (B) In the discrete metric $d(x, y) = 1$ for $x \neq y$ and $d(x, x) = 0$. So $B(x_0, 1) = \{x : d(x, x_0) < 1\} = \{x_0\}$, since no other point has distance < 1 from x_0.

Q3. A point x belongs to the *frontier* (boundary) ∂A of a set A if and only if

(A) Every open ball around x lies entirely in A

(B) Every open ball around x lies entirely in A^c

(C) Every open ball around x intersects both A and A^c

(D) $x \in A$ and $x \notin \overline{A^c}$

Answer: (C) $x \in \partial A \Leftrightarrow$ every neighbourhood (open ball) of x contains at least one point of A and at least one point of A^c. Equivalently $x \in \bar{A} \cap \overline{A^c}$.

Q4. In $(\mathbb{R}, d)$ with the usual metric, the interior of $[a, b]$ is

(A) $[a, b]$

(B) (a, b)

(C) $\{a, b\}$

(D) $\emptyset$

Answer: (B) The interior consists of all points x for which some open ball $B(x, r) \subseteq [a, b]$. This holds precisely for $x \in (a, b)$. The endpoints a and b have every neighbourhood containing points outside $[a, b]$, so they are not interior points.

Q5. A sequence $\{x_n\}$ in a metric space (X, d) converges to L if

(A) $d(x_n, L) \to \infty$ as $n \to \infty$

(B) $d(x_n, L) \to 0$ as $n \to \infty$

(C) $x_n = L$ for all n

(D) $d(x_n, x_m) \to L$ for all m, n

Answer: (B) $x_n \to L$ in (X, d) iff $\forall\, \varepsilon > 0, \exists\, N$ such that $n \geq N \Rightarrow d(x_n, L) < \varepsilon$. Equivalently, $\lim_{n\to\infty} d(x_n, L) = 0$.

Q6. In the metric space $(\mathbb{R}^2, d_\infty)$ with $d_\infty(x, y) = \max(|x_1 - y_1|, |x_2 - y_2|)$, the open ball $B(\mathbf{0}, 1)$ is

(A) A disk of radius 1

(B) An open square $(-1, 1) \times (-1, 1)$

(C) A diamond (rhombus)

(D) The whole plane $\mathbb{R}^2$

Answer: (B) $B(\mathbf{0}, 1) = \{(x_1, x_2) : \max(|x_1|, |x_2|) < 1\} = (-1, 1) \times (-1, 1)$, which is an **open square**. The Euclidean ball is a disk; the taxicab ball is a diamond.

In PSC exams, the three most tested metrics are Euclidean (d_2), Taxicab (d_1), and Chebyshev (d_∞). Their unit balls are: **circle**, **diamond (rhombus)**, and **square** respectively. Memorise these shapes!

2. Topological Spaces

A **topological space** (X, τ) is a set X with a collection τ of subsets (called **open sets**) satisfying:

1. $\emptyset \in \tau$ and $X \in \tau$
2. Arbitrary unions of members of τ belong to τ
3. Finite intersections of members of τ belong to τ

Examples:

- **Indiscrete topology:** $\tau = \{\emptyset, X\}$ — coarsest topology.
- **Discrete topology:** $\tau = 2^X$ (all subsets open) — finest topology.
- **Metric topology:** open sets are unions of open balls.

A set is **closed** if its complement is open. The **closure** $\bar{A}$ is the smallest closed set containing A.

2.1. MCQs — Topological Spaces

Q7. Which of the following is a valid topology on $X = \{a, b, c\}$?

(A) $\tau = \{\emptyset, \{a\}, \{b\}, X\}$

(B) $\tau = \{\emptyset, \{a, b\}, \{b, c\}, X\}$

(C) $\tau = \{\emptyset, \{a\}, \{a, b\}, X\}$

(D) $\tau = \{\{a\}, \{b\}, X\}$

Answer: (C) Check axioms for $\tau = \{\emptyset, \{a\}, \{a, b\}, X\}$: (i) $\emptyset, X \in \tau$ ✓; (ii) $\{a\} \cup \{a, b\} = \{a, b\} \in \tau$ ✓; (iii) $\{a\} \cap \{a, b\} = \{a\} \in \tau$ ✓. Valid topology. Option (A) fails: $\{a\} \cup \{b\} = \{a, b\} \notin \tau$. Option (D) fails: $\emptyset \notin \tau$.

Q8. The indiscrete topology on a set X consists of

(A) All subsets of X

(B) Only $\emptyset$ and X

(C) All singletons plus X

(D) All finite subsets of X

Answer: (B) The **indiscrete** (or trivial) topology is $\tau = \{\emptyset, X\}$, containing only the empty set and the whole set. It is the coarsest possible topology.

Q9. In any topological space, the intersection of *finitely many* open sets is

(A) Always open

(B) Always closed

(C) Not necessarily open

(D) Always empty

Answer: (A) By definition of topology, **finite intersections** of open sets are open. Note: *infinite* intersections of open sets need not be open (e.g. $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$ is closed in $\mathbb{R}$).

Q10. A subset A of a topological space (X, τ) is closed if and only if

(A) $A \in \tau$

(B) $A^c \in \tau$

(C) $A = \bar{A}$

(D) Both (B) and (C)

Answer: (D) Both definitions are equivalent: A is closed $\Leftrightarrow$ A^c is open $\Leftrightarrow$ A contains all its limit points $\Leftrightarrow$ $A = \bar{A}$.

A set can be both open and closed ("clopen") — e.g. $\emptyset$ and X itself are always clopen. In a **discrete** space, *every* set is clopen.

3. Normed Spaces & Inner Product Spaces

A **normed space** $(V, \|\cdot\|)$ is a vector space V over $\mathbb{F}$ with a norm satisfying:

1. $\|x\| \geq 0$, $\|x\| = 0 \Leftrightarrow x = \mathbf{0}$
2. $\|\alpha x\| = |\alpha| \, \|x\|$
3. $\|x + y\| \leq \|x\| + \|y\|$ (Triangle inequality)

A **Banach space** is a *complete* normed space (every Cauchy sequence converges).

An **inner product space** $(V, \langle \cdot, \cdot \rangle)$ has an inner product satisfying:

1. $\langle x, x \rangle \geq 0$, $\langle x, x \rangle = 0 \Leftrightarrow x = \mathbf{0}$
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$ (conjugate symmetry)
3. $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$ (linearity)

Every inner product induces a norm: $\|x\| = \sqrt{\langle x, x \rangle}$.
A **Hilbert space** is a complete inner product space.

Key Formulas & Results

$$\text{Cauchy–Schwarz Inequality: } |\langle x, y \rangle| \leq \|x\| \, \|y\|$$

$$\text{Parallelogram Law: } \|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Standard inner product in $\mathbb{R}^n$: $\langle x, y \rangle = \sum_{k=1}^{n} x_k y_k$ In L^2: $\langle f, g \rangle = \int f(x)\overline{g(x)} \, dx$

3.1. MCQs — Normed & Inner Product Spaces

Q11. The norm $\|x\|_1 = \sum_{k=1}^{n} |x_k|$ on $\mathbb{R}^n$ is called the

(A) Euclidean norm

(B) Taxicab (Manhattan) norm

(C) Chebyshev norm

(D) Frobenius norm

Answer: (B) $\|x\|_1 = \sum |x_k|$ is the ℓ^1 **(Taxicab or Manhattan) norm**. The Euclidean norm is $\|x\|_2 = \sqrt{\sum x_k^2}$; the Chebyshev norm is $\|x\|_\infty = \max |x_k|$.

Q12. The Cauchy–Schwarz inequality states that for vectors x, y in an inner product space

(A) $|\langle x, y \rangle| \geq \|x\| \, \|y\|$

(B) $|\langle x, y\rangle| = \|x\|\,\|y\|$

(C) $|\langle x, y\rangle| \le \|x\| + \|y\|$

(D) $|\langle x, y\rangle| \le \|x\|\,\|y\|$

Answer: (D) The **Cauchy–Schwarz inequality**: $|\langle x, y\rangle| \le \|x\|\,\|y\|$. Equality holds if and only if x and y are linearly dependent (one is a scalar multiple of the other).

Q13. Which of the following satisfies the parallelogram law and is therefore induced by an inner product?

(A) $\|x\|_1$ on $\mathbb{R}^2$

(B) $\|x\|_\infty$ on $\mathbb{R}^2$

(C) $\|x\|_2$ on $\mathbb{R}^2$

(D) $\|x\|_p$ for $p \neq 2$

Answer: (C) A norm comes from an inner product if and only if it satisfies the **Parallelogram Law**: $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$. Only $\|\cdot\|_2$ (Euclidean) satisfies this; ℓ^p norms for $p \neq 2$ do not.

Q14. A Hilbert space is defined as

(A) A normed space that is complete

(B) A vector space with an inner product

(C) A complete inner product space

(D) A finite-dimensional inner product space

Answer: (C) A **Hilbert space** is a *complete* inner product space. Completeness means every Cauchy sequence converges within the space. A complete normed space (without requiring inner product structure) is called a Banach space.

Q15. Two vectors u and v in an inner product space are orthogonal if

(A) $\|u\| = \|v\|$

(B) $\langle u, v\rangle = 0$

(C) $u + v = \mathbf{0}$

(D) $\|u+v\| = \|u\| + \|v\|$

Answer: (B) $u \perp v \Leftrightarrow \langle u, v\rangle = 0$. By the Pythagorean theorem in inner product spaces: if $\langle u, v\rangle = 0$ then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.

4. Gram–Schmidt Orthonormalization Process

Given a linearly independent set $\{v_1, v_2, \ldots, v_n\}$ in an inner product space, the **Gram–Schmidt process** produces an orthonormal set $\{e_1, e_2, \ldots, e_n\}$ spanning the same space:

$$u_1 = v_1$$
$$u_2 = v_2 - \frac{\langle v_2,\, u_1\rangle}{\langle u_1,\, u_1\rangle} u_1$$
$$u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k,\, u_j\rangle}{\langle u_j,\, u_j\rangle} u_j$$

Then normalise: $e_k = \dfrac{u_k}{\|u_k\|}$.

Key Formulas & Results

$$\text{Projection of } v \text{ onto } u: \quad \text{proj}_u v = \frac{\langle v,\, u\rangle}{\langle u,\, u\rangle} u = \frac{\langle v,\, u\rangle}{\|u\|^2} u$$

$$\text{Orthonormal basis: } \langle e_i,\, e_j\rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

4.1. MCQs — Gram–Schmidt Process

Q16. Apply Gram–Schmidt to $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$. The second orthonormal vector e_2 is

(A) $(1, 1, 0)/\sqrt{2}$

(B) $(0, 1, 0)$

(C) $(1, 0, 0)$

(D) $(0, 0, 1)$

Answer: (B) $e_1 = v_1 = (1, 0, 0)$.
$u_2 = v_2 - \langle v_2,\, e_1\rangle\, e_1 = (1, 1, 0) - (1)(1, 0, 0) = (0, 1, 0)$. Since $\|u_2\| = 1$, we get $e_2 = u_2 = (0, 1, 0)$.

Q17. The projection of $v = (3, 4)$ onto $u = (1, 0)$ is

(A) $(4, 0)$

(B) $(3, 0)$

(C) $(3, 4)$

(D) $(0, 4)$

Answer: (B) $\text{proj}_u v = \dfrac{\langle v, u\rangle}{\|u\|^2} u = \dfrac{3}{1}(1,0) = (3,0).$

Q18. The Gram–Schmidt process converts a linearly independent set into

(A) A linearly dependent set

(B) An orthonormal set spanning the same subspace

(C) An orthonormal set spanning a larger subspace

(D) A set of unit vectors that may not be orthogonal

Answer: (B) The Gram–Schmidt process produces an **orthonormal set** that spans exactly the same subspace as the original linearly independent set. No vectors are added or removed; only their directions are adjusted.

Q19. After applying the Gram–Schmidt process, the resulting vectors $\{e_1, e_2, \ldots, e_n\}$ satisfy

(A) $\langle e_i, e_j\rangle = 1$ for all i, j

(B) $\langle e_i, e_j\rangle = \delta_{ij}$ (Kronecker delta)

(C) $\langle e_i, e_j\rangle = 0$ for all i, j

(D) $\|e_i\| = i$ for each i

Answer: (B) After Gram–Schmidt normalisation, $\langle e_i, e_j\rangle = \delta_{ij}$: each vector has unit length ($\langle e_i, e_i\rangle = 1$) and distinct vectors are orthogonal ($\langle e_i, e_j\rangle = 0$ for $i \neq j$).

In PSC questions on Gram–Schmidt: always subtract the full projection onto *each* previous vector before normalising. A common mistake is projecting onto only the immediately preceding vector.

5. Matrices, Echelon Forms & Linear Transformations

Row Reduction & Echelon Forms

Row Echelon Form (REF): Each leading entry (pivot) is to the right of the pivot in the row above; all zero rows are at the bottom.

Reduced Row Echelon Form (RREF): REF plus: each pivot is 1 and is the only nonzero entry in its column.

Elementary row operations:

- $R_i \leftrightarrow R_j$ (swap rows)
- $R_i \leftarrow cR_i$, $c \neq 0$ (scale a row)
- $R_i \leftarrow R_i + cR_j$ (add multiple of one row to another)

A system $A\mathbf{x} = \mathbf{b}$ is solved by row-reducing the augmented matrix $[A \mid \mathbf{b}]$.

A function $T : V \to W$ between vector spaces is a **linear transformation** if:

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v) \quad \forall\, u, v \in V,\ \alpha, \beta \in \mathbb{F}$$

Kernel (null space): $\ker T = \{v \in V : T(v) = \mathbf{0}\}$
Range (image): $\operatorname{Im} T = \{T(v) : v \in V\}$
Rank–Nullity Theorem: $\dim(\ker T) + \dim(\operatorname{Im} T) = \dim V$

Key Formulas & Results

$$\text{Rank–Nullity: } \operatorname{rank}(A) + \operatorname{nullity}(A) = n \quad (A \text{ is } m \times n)$$

$$\text{System } A\mathbf{x} = \mathbf{b}: \quad \text{consistent iff } \operatorname{rank}(A) = \operatorname{rank}([A|\mathbf{b}])$$

Unique solution iff $\operatorname{rank}(A) = n$; $A\mathbf{x} = \mathbf{0}$ has non-trivial solution iff $\operatorname{rank}(A) < n$

5.1. MCQs — Matrices & Linear Transformations

Q20. The rank of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$ is

(A) 3
(B) 2
(C) 1
(D) 0

Answer: (C) Rows 2 and 3 are multiples of Row 1 ($R_2 = 2R_1$, $R_3 = 3R_1$). After row reduction only one non-zero row remains, so $\operatorname{rank}(A) = \mathbf{1}$.

Q21. If $T : \mathbb{R}^3 \to \mathbb{R}^4$ is a linear transformation with $\dim(\ker T) = 1$, then $\dim(\operatorname{Im} T)$ is

(A) 4
(B) 3
(C) 2
(D) 1

Answer: (C) By the Rank–Nullity Theorem: $\dim(\ker T) + \dim(\operatorname{Im} T) = \dim(\mathbb{R}^3) = 3$. So $\dim(\operatorname{Im} T) = 3 - 1 = \mathbf{2}$.

Q22. The system $A\mathbf{x} = \mathbf{0}$ (homogeneous) always has

(A) Only the trivial solution $\mathbf{x} = \mathbf{0}$

(B) No solution

(C) At least the trivial solution $\mathbf{x} = \mathbf{0}$

(D) Infinitely many solutions always

Answer: (C) A homogeneous system $A\mathbf{x} = \mathbf{0}$ *always* has at least the **trivial solution** $\mathbf{x} = \mathbf{0}$. It has infinitely many solutions (non-trivial ones) iff $\operatorname{rank}(A) < n$ (number of unknowns), i.e. iff $\det(A) = 0$ for a square matrix.

Q23. The reduced row echelon form (RREF) of $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$ is

(A) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$

(C) $\begin{pmatrix} 2 & 4 \\ 0 & 0 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$

Answer: (B) $R_1 \leftarrow R_1/2$: $\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$. Then $R_2 \leftarrow R_2 - R_1$: $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$. This is already in RREF: pivot in column 1 is 1, no other entries above/below, and row 2 is all zeros.

Q24. A linear transformation $T : V \to W$ is injective (one-to-one) if and only if

(A) $\dim(\operatorname{Im} T) = \dim W$

(B) $\ker T = \{\mathbf{0}\}$

(C) T is surjective

(D) $\dim V = \dim W$

Answer: (B) T is injective $\Leftrightarrow \ker T = \{\mathbf{0}\} \Leftrightarrow \operatorname{nullity}(T) = 0$. This means different inputs give different outputs: $T(u) = T(v) \Rightarrow u = v$.

6. Systems of Equations: Gauss–Seidel & Jacobi Methods

For the system $A\mathbf{x} = \mathbf{b}$ with A diagonally dominant, iterative methods converge. Write $A = D + L + U$ where D = diagonal, L = strictly lower triangular, U = strictly upper triangular.

Jacobi Method: Use values from the *previous* iteration only:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right)$$

Gauss–Seidel Method: Use the *most recently updated* values:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right)$$

Gauss–Seidel converges faster than Jacobi (generally about twice as fast) for diagonally dominant systems.

Key Formulas & Results

Diagonal Dominance (sufficient condition for convergence):

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \text{for all } i$$

Jacobi matrix: $B_J = -D^{-1}(L + U)$
Gauss–Seidel matrix: $B_{GS} = -(D + L)^{-1}U$

6.1. MCQs — Iterative Methods

Q25. The key difference between Gauss–Seidel and Jacobi methods is that Gauss–Seidel

(A) Uses values only from the previous iteration

(B) Uses the most recently computed values within the same iteration

(C) Requires a symmetric matrix

(D) Is always slower than Jacobi

Answer: (B) In **Gauss–Seidel**, as soon as $x_i^{(k+1)}$ is computed, it is immediately used in computing $x_{i+1}^{(k+1)}$. This use of updated values typically makes Gauss–Seidel converge about *twice* as fast as Jacobi.

Q26. For the system $10x + y = 11$, $x + 10y = 11$, one Jacobi iteration starting from $(0, 0)$ gives

(A) $x^{(1)} = 1.1,\ y^{(1)} = 1.1$

(B) $x^{(1)} = 1.1,\ y^{(1)} = 1.0$

(C) $x^{(1)} = 1.0,\ y^{(1)} = 1.1$

(D) $x^{(1)} = 0.1,\ y^{(1)} = 0.1$

Answer: (A) Jacobi formulas: $x^{(1)} = \dfrac{11 - y^{(0)}}{10} = \dfrac{11 - 0}{10} = 1.1$; $y^{(1)} = \dfrac{11 - x^{(0)}}{10} = \dfrac{11 - 0}{10} = 1.1$. (Exact solution: $x = y = 1$. Convergence is guaranteed since the matrix is diagonally dominant: $|10| > |1|$.)

Q27. A sufficient condition for both Jacobi and Gauss–Seidel methods to converge is that A is

(A) Symmetric

(B) Positive definite

(C) Strictly diagonally dominant

(D) Upper triangular

Answer: (C) **Strict diagonal dominance** ($|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all i) guarantees convergence of both iterative methods. Positive definiteness guarantees Gauss–Seidel convergence but not necessarily Jacobi.

Q28. In one Gauss–Seidel iteration for $4x - y = 7$, $-x + 4y = 7$, starting from $(0, 0)$:

(A) $x^{(1)} = 1.75,\ y^{(1)} = 1.75$

(B) $x^{(1)} = 1.75,\ y^{(1)} = 2.1875$

(C) $x^{(1)} = 2,\ y^{(1)} = 2$

(D) $x^{(1)} = 1,\ y^{(1)} = 2$

Answer: (B) GS uses updated values immediately:

$$x^{(1)} = \frac{7 + y^{(0)}}{4} = \frac{7 + 0}{4} = 1.75$$

$$y^{(1)} = \frac{7 + x^{(1)}}{4} = \frac{7 + 1.75}{4} = \frac{8.75}{4} = 2.1875$$

(Exact solution: $x = y = 7/3 \approx 2.333$.)

Key PSC distinction: **Jacobi** updates all variables *simultaneously* using old values; **Gauss–Seidel** updates *sequentially* using the latest available values. This makes Gauss–Seidel generally faster but requires sequential computation.

7. Determinants — Properties & Computation

For an $n \times n$ matrix A:

1. $\det(A) = \det(A^T)$
2. $\det(AB) = \det(A)\det(B)$
3. If two rows (or columns) are identical, $\det(A) = 0$
4. Swapping two rows changes the sign: $\det \to -\det$
5. Multiplying a row by scalar c multiplies det by c
6. Adding a multiple of one row to another does not change det
7. $\det(cA) = c^n \det(A)$ for an $n \times n$ matrix
8. A is invertible $\Leftrightarrow \det(A) \neq 0$
9. $\det(A^{-1}) = 1/\det(A)$

Key Formulas & Results

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

$$\text{Cofactor expansion along row } i: \quad \det(A) = \sum_{j=1}^{n} a_{ij}(-1)^{i+j} M_{ij}$$

$$\text{Cramer's Rule: } x_i = \frac{\det(A_i)}{\det(A)}, \quad \det(A) \neq 0$$

7.1. MCQs — Determinants

Q29. If $\det(A) = 5$ for a 3×3 matrix, then $\det(2A)$ equals

(A) 10

(B) 20

(C) 40

(D) 80

Answer: (C) $\det(cA) = c^n \det(A)$. For $n = 3$: $\det(2A) = 2^3 \cdot \det(A) = 8 \times 5 =$ **40**.

Q30. If matrix A has two identical rows, then

(A) $\det(A) = 1$

(B) $\det(A) = -1$

(C) $\det(A) = 0$

(D) $\det(A)$ depends on the rows

Answer: (C) Swapping two identical rows changes the sign of the determinant: $\det \to -\det$. But swapping identical rows leaves the matrix unchanged: $\det = \det$. Thus $\det = -\det$, giving $\det(A) = \mathbf{0}$.

Q31. $\det(AB)$ where $\det(A) = 3$ and $\det(B) = 4$ equals

(A) 7

(B) 12

(C) 1

(D) 81

Answer: (B) The **multiplicative property**: $\det(AB) = \det(A) \cdot \det(B) = 3 \times 4 = \mathbf{12}$.

Q32. The matrix $A = \begin{pmatrix} 3 & 1 \\ 6 & 2 \end{pmatrix}$ is singular because

(A) Its trace is zero

(B) $\det(A) = 0$

(C) Its rows are not integers

(D) $\det(A) = 12$

Answer: (B) $\det(A) = 3 \cdot 2 - 1 \cdot 6 = 6 - 6 = 0$. Since $\det(A) = 0$, A is **singular** (not invertible). Note: Row 2 $= 2\times$ Row 1, confirming linear dependence.

Q33. If A is an invertible matrix, then $\det(A^{-1})$ equals

(A) $-\det(A)$

(B) $\det(A)$

(C) $1/\det(A)$

(D) $\det(A)^2$

Answer: (C) From $AA^{-1} = I$: $\det(A)\det(A^{-1}) = \det(I) = 1$, so $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

8. Eigenvalues, Eigenvectors & Diagonalization

For an $n \times n$ matrix A, a scalar λ is an **eigenvalue** and a non-zero vector $\mathbf{x}$ is the corresponding **eigenvector** if:

$$A\mathbf{x} = \lambda\mathbf{x} \quad \Leftrightarrow \quad (A - \lambda I)\mathbf{x} = \mathbf{0}$$

Characteristic equation: $\det(A - \lambda I) = 0$ (a degree-n polynomial in λ).

Diagonalization: A is diagonalizable if it has n linearly independent eigenvectors. Then:

$$A = PDP^{-1}$$

where $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and $P = [\mathbf{x}_1 \mid \cdots \mid \mathbf{x}_n]$ (eigenvectors as columns).

Symmetric matrices ($A = A^T$) are always diagonalizable with real eigenvalues and orthogonal eigenvectors from distinct eigenvalues. Hence $A = PDP^T$ with P orthogonal.

Key Formulas & Results

$$\text{Characteristic polynomial: } p(\lambda) = \det(A - \lambda I)$$

$$\text{Trace} = \lambda_1 + \lambda_2 + \cdots + \lambda_n = \sum a_{ii}, \qquad \det(A) = \lambda_1 \cdot \lambda_2 \cdots \lambda_n$$

$$\text{Spectral Theorem (symmetric): } A = PDP^T, \quad P^{-1} = P^T$$

8.1. MCQs — Eigenvalues & Eigenvectors

Q34. The eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ are

(A) $\lambda_1 = 5,\ \lambda_2 = 2$

(B) $\lambda_1 = 4,\ \lambda_2 = 3$

(C) $\lambda_1 = 7,\ \lambda_2 = 0$

(D) $\lambda_1 = 3,\ \lambda_2 = 2$

Answer: (A) $\det(A-\lambda I) = (4-\lambda)(3-\lambda)-2 = \lambda^2-7\lambda+10 = (\lambda-5)(\lambda-2) = 0$. So $\lambda_1 = 5$, $\lambda_2 = 2$. Check: trace $= 4+3 = 7 = 5+2$ ✓; det $= 12-2 = 10 = 5\times 2$ ✓.

Q35. An eigenvector corresponding to $\lambda = 5$ of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ is

(A) $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$

(D) $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$

Answer: (B) $(A-5I)\mathbf{x} = \mathbf{0}$: $\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}\mathbf{x} = \mathbf{0}$. Row reduce: $-x_1 + x_2 = 0 \Rightarrow x_1 = x_2$. So eigenvector is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Q36. A real symmetric matrix always has

(A) Complex eigenvalues

(B) Zero eigenvalues

(C) Real eigenvalues and orthogonal eigenvectors for distinct eigenvalues

(D) Repeated eigenvalues

Answer: (C) By the **Spectral Theorem**: every real symmetric matrix has (i) all *real* eigenvalues and (ii) eigenvectors corresponding to *distinct* eigenvalues are mutually orthogonal. It is always orthogonally diagonalizable: $A = PDP^T$.

Q37. A matrix A is diagonalizable if and only if

(A) $\det(A) \neq 0$

(B) A is symmetric

(C) A has n linearly independent eigenvectors

(D) All eigenvalues are distinct

Answer: (C) A is diagonalizable $\Leftrightarrow$ it has n linearly independent eigenvectors (forming the columns of P). Note: distinct eigenvalues guarantee this (sufficient), but it is not necessary — a matrix can have repeated eigenvalues and still be diagonalizable.

Q38. The eigenvalues of a triangular matrix are

(A) The entries of the last column

(B) The entries on the main diagonal

(C) The square roots of the diagonal entries

(D) Always 0 and 1

Answer: (B) For a triangular (upper or lower) matrix, $\det(A-\lambda I) = \prod_{i=1}^{n}(a_{ii}-\lambda)$. Hence the eigenvalues are precisely the **main diagonal entries** $a_{11}, a_{22}, \ldots, a_{nn}$.

Q39. If λ is an eigenvalue of A, then λ^2 is an eigenvalue of

(A) $2A$

(B) A^2

(C) $A+I$

(D) A^T

Answer: (B) If $A\mathbf{x} = \lambda\mathbf{x}$, then $A^2\mathbf{x} = A(A\mathbf{x}) = A(\lambda\mathbf{x}) = \lambda(A\mathbf{x}) = \lambda^2\mathbf{x}$. So λ^2 is an eigenvalue of $\mathbf{A^2}$.

Q40. The diagonalization $A = PDP^{-1}$ gives A^k equal to

(A) PD^kP^{-1}

(B) P^kDQ^{-k}

(C) $kPDP^{-1}$

(D) PD^kP

Answer: (A) $A^2 = PDP^{-1} \cdot PDP^{-1} = PD^2P^{-1}$. By induction, $A^k = PD^kP^{-1}$. Since D is diagonal, $D^k = \mathrm{diag}(\lambda_1^k, \ldots, \lambda_n^k)$, making powers easy to compute.

For PSC exams: always find eigenvalues from $\det(A - \lambda I) = 0$ first, then find eigenvectors by solving $(A - \lambda I)\mathbf{x} = \mathbf{0}$. For a 2×2 matrix: eigenvalues satisfy $\lambda^2 - (\mathrm{trace})\lambda + \det = 0$.

9. Additional Mixed MCQs

9.1. Metric & Functional Analysis — Mixed

Q41. A metric space (X, d) is said to be *complete* if
- (A) Every sequence converges
- (B) Every bounded sequence converges
- (C) Every Cauchy sequence in X converges to a point in X
- (D) X is finite

Answer: (C) Completeness: every Cauchy sequence in X has its limit inside X. $\mathbb{R}$ with the usual metric is complete; $\mathbb{Q}$ is not (the sequence of rational approximations to $\sqrt{2}$ is Cauchy in $\mathbb{Q}$ but converges to $\sqrt{2} \notin \mathbb{Q}$).

Q42. Every open set in a metric space is
- (A) A union of closed balls
- (B) A union of open balls
- (C) A single open ball
- (D) Closed

Answer: (B) In the metric topology, every open set is *by definition* a union of open balls (the basic open sets). This is the standard basis for the metric topology.

Q43. The space ℓ^2 of square-summable sequences is an example of
- (A) A Banach space that is not a Hilbert space
- (B) A Hilbert space with inner product $\langle x, y\rangle = \sum x_n \bar{y}_n$
- (C) A metric space that is not a normed space
- (D) A finite-dimensional vector space

Answer: (B) $\ell^2 = \{(x_n) : \sum |x_n|^2 < \infty\}$ with $\langle x, y\rangle = \sum x_n \bar{y}_n$ is a **Hilbert space**: the inner product is well-defined, the induced norm is complete, and it is infinite-dimensional.

9.2. Linear Algebra — Mixed

Q44. For two $n \times n$ matrices A and B, which equality is always true?
- (A) $AB = BA$
- (B) $(AB)^T = A^T B^T$

(C) $(AB)^T = B^T A^T$

(D) $\det(A + B) = \det(A) + \det(B)$

Answer: (C) $(AB)^T = B^T A^T$ (reverse order). Option (A) fails in general (matrix multiplication is not commutative). Option (D) is false: determinant is not additive.

Q45. The set of all solutions to $A\mathbf{x} = \mathbf{0}$ is called the

(A) Column space of A

(B) Row space of A

(C) Null space (kernel) of A

(D) Range of A

Answer: (C) The solution set of the homogeneous system $A\mathbf{x} = \mathbf{0}$ is the **null space** (or kernel) of A: $\ker A = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$. Its dimension is the nullity of A.

Q46. If A is a 4×5 matrix with $\text{rank}(A) = 3$, then the nullity of A is

(A) 1

(B) 2

(C) 3

(D) 4

Answer: (B) Rank–Nullity Theorem: $\text{rank}(A) + \text{nullity}(A) = n = 5$ (number of columns). So $\text{nullity}(A) = 5 - 3 = \mathbf{2}$.

Q47. An $n \times n$ matrix A is orthogonal if and only if

(A) $A^T = A$

(B) $A^T = -A$

(C) $A^T A = AA^T = I$

(D) $\det(A) = 0$

Answer: (C) A matrix is **orthogonal** if $A^T A = I$ (equivalently $A^{-1} = A^T$). Consequences: $\det(A) = \pm 1$ and all eigenvalues have modulus 1. Orthogonal matrices represent rotations and reflections.

10. Quick Summary Tables

10.1. Common Metrics and Their Unit Balls in $\mathbb{R}^2$

Name	Formula	Unit Ball Shape	Completeness
Euclidean (d_2)	$\sqrt{(x_1-y_1)^2+(x_2-y_2)^2}$	Circle	Yes
Taxicab (d_1)	$\lvert x_1-y_1\rvert+\lvert x_2-y_2\rvert$	Diamond (Rhombus)	Yes
Chebyshev (d_∞)	$\max(\lvert x_1-y_1\rvert,\lvert x_2-y_2\rvert)$	Square	Yes
Discrete	0 if $x=y$; 1 otherwise	$\{x_0\}$	Yes

10.2. Jacobi vs Gauss–Seidel Comparison

Feature	Jacobi Method	Gauss–Seidel Method
Value used	Previous iteration only	Most recently updated
Update order	Simultaneous	Sequential
Storage	Two arrays needed	One array suffices
Convergence speed	Slower	Faster (generally $\sim 2\times$)
Convergence condition	Strict diagonal dominance	Same (or positive definite)
Matrix form	$B_J=-D^{-1}(L+U)$	$B_{GS}=-(D+L)^{-1}U$

10.3. Eigenvalue Summary for Special Matrices

Matrix Type	Eigenvalue Property	Diagonalizable?
Symmetric ($A=A^T$)	All real	Always (orthogonally)
Skew-symmetric ($A=-A^T$)	Pure imaginary or 0	Over $\mathbb{C}$
Orthogonal ($A^TA=I$)	$\lvert\lambda\rvert=1$	Over $\mathbb{C}$
Triangular	Diagonal entries	If no repeated eigenvalues
Idempotent ($A^2=A$)	Only 0 or 1	Always
Nilpotent ($A^k=0$)	All zero	Only if $A=0$

10.4. Key Functional Analysis Spaces

Space	Definition	Norm/Inner Product	Complete?
$\mathbb{R}^n$	n-tuples of reals	$\|x\|_2 = \sqrt{\sum x_i^2}$	Yes (Banach)
ℓ^p $(1 \leq p < \infty)$	$\sum \|x_n\|^p < \infty$	$\|x\|_p = (\sum \|x_n\|^p)^{1/p}$	Yes (Banach)
ℓ^2	$\sum \|x_n\|^2 < \infty$	$\langle x, y\rangle = \sum x_n \bar{y}_n$	Yes (Hilbert)
$C[a, b]$	Continuous functions	$\|f\|_\infty = \sup \|f(x)\|$	Yes (Banach)
$L^2[a, b]$	Square-integrable	$\langle f, g\rangle = \int_a^b f\bar{g}\, dx$	Yes (Hilbert)

End of Metric Spaces, Functional Analysis, Linear Algebra & Numerical Analysis
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